



Perimeter, Area & Algebraic Expressions

Day 16 of 20 · Theme: Generic Amusement Park

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.7

Standard: 9.3.5.8

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can apply the lesson's core skill with fluency and check my answers for reasonableness.
- I can explain my reasoning using math vocabulary.
- I can connect the math to the lesson's real-world career and cultural context.

ENGAGEMENT & PACING

Strategy: Collaborative Whiteboard **Practice Format:** Large Group Practice **Duration:** 50 minutes

TEACHER ANNOTATIONS — PRACTICE ANSWERS (VERIFIED)

Scaffold (Tier 1) Prompt: A rectangle's length is 3 more than its width w . Write an expression for the perimeter. *Model answers:*

• $4w + 6$ **On-Level (Tier 2) Prompt:** A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter. *Model answers:*

• $A = s^2$, $P = 4s$

• $4x + 5$ **Extension (Tier 3) Prompt:** A rectangle has area 48 and width 6. Write and solve an equation for the length l . *Model answers:*

• $6l = 48$, $l = 8$

EXIT TICKET — ANSWER & SCORING

Exit ticket: Show your work for full credit *Scoring: 1 point per exit ticket; mastery = 4 of 4. Collect before the final 5 minutes; address misconceptions next lesson's bell ringer.*

DIFFERENTIATION (3 TIERS)

- Tier 1 (Support): Use tiered practice 'go'; provide a GEMDAS keyword bank and sentence starters.
- Tier 2 (On-Level): Work 'set' problems; partner check before whole-class reveal.
- Tier 3 (Challenge): Complete 'extra' extension; ask students to create their own verbal phrase.

CULTURE & CAREER CONNECTION

- Culture: Sami (Sápmi / Northern Europe) The Sámi are the Indigenous people of Sápmi, spanning northern Norway, Sweden, Finland, and Russia. Sami culture centers on reindeer herding and duodji (handicraft).

- Career: Andersen Corp — Bayport | Window Energy Rater | AA | \$80,142 O*NET SOC: 47-4011.00

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